MILITARY SPECIFICATION

BALLISTICS COMPUTER, SUBASSEMBLY: 10543700 (COMMANDER'S CONTROL UNIT)

1. SCOPE

1.1 This specification covers one type of electro-mechanical assembly known as Ballistics Computer, Subassembly: 10543700 as (Commander's Control Unit) (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel: Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army, Frankford Arsenal

F10543700	Ballistics Computer, Subassembly: 10543700 (Commander's Control Unit)

PACKAGING DATA SHEET

MIL-P-14232/P10543700	Packaging of Commander's Control Unit 10543700

FSC 1220

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(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The Commander's Control Unit shall be manufactured in accordance with Drawing F10543700 and drawings pertaining thereto.

3.2 General specification.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of MIL-F-13926:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

3.3 Environmental.

3.3.1 Storage temperatures.- There shall be no evidence of physical failures such as glass breakage or casting failures due to storage temperatures ranging from -80° to +160°F. (Fahrenheit).

3.3.2 Operating temperatures.- The Commander's Control Unit shall meet all the requirements of 3.5.2 through 3.5.9 inclusive, while exposed to and thermally stabilized at ambient temperatures of -40° and +150°F.

3.3.3 Vibration.- The slippable scales on the boresight knobs of the Commander's Control Unit shall be movable under normal finger pressure, but shall not move under vibration. The unit shall meet the requirements of 3.4.1 through 3.5.9 after being vibrated in a vertical plane for five minutes ± 15 seconds at an amplitude of 1/16 inch (1/8 inch total excursion) at a frequency of 30 cycles per second.

3.3.4 Water resistance (sealing).- The interior of the unit shall show no evidence of water damage after being subjected to a simulated rainfall.

3.4 Electrical.

3.4.1 Continuity.- The continuity of the unit shall conform to Drawing F10543700.

3.5 Performance.

3.5.1 Input power.- The unit shall meet the requirements of 3.5.2 through 3.5.9 when excited with zero and Pi phase, 51. ± 0.5 volts

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P to P, 400 CPS ± 1%, connected across pins V and e of 2J1. Pin W of 2J2 shall be connected to pin Z of 2J2. The A.C. common shall be connected to pin r of 2J1. Harmonic distortion shall not exceed 2%. Variation in harmonic distortion shall not exceed 0.5% during tests. Source impedance of the A.C. supply shall be less than 0.10 ohm.

3.5.2 Power supply output.- With the A.C. supply, as specified in 3.5.1 and a 0.25 ampere ± 1% load connected across pins F and W of 2J2, the D.C. voltage at pin F shall be 23.0 volts ± 5%. The ripple across the load shall have a maximum true RMS value of 0.45 volt. The ripple frequency shall be twice that of the A.C. supply.

3.5.3 Switch lens illumination.- With the A.C. supply as specified in 3.5.1, the four ammunition selector switch lenses shall be illuminated. Alternate connection of pins E, F, G, and H of 2J1 to 2J2-a shall increase the intensity of the red, blue, yellow and green lenses respectively. Rheostat 2R1 shall vary the illumination of the switch lenses.

3.5.4 Control indicator lamps.- When pin z of 2J1 is connected to ground through 100 ohms, the red and green control lamps shall alternately be illuminated and extinguished as the override switch is actuated. The red lamp shall be illuminated when the ground connection at pin z of 2J1 is removed. Rheostat 2R1 shall vary the intensity of the control indicator lamps.

3.5.5 Switch resistance.- The maximum allowable switch resistance shall be as indicated in Table I.

TABLE I

SWITCH	RESISTANCE	MEASURED BETWEEN PINS	
2S1	.2 ohm	2J1-A	2J1-a
2S2	.2 ohm	2J1-B	2J1-a
2S3	.2 ohm	2J1-C	2J1-a
2S4	.2 ohm	2J1-D	2J1-a
2S5	.2 ohm	2J1-j	2J2-F
2S6	.2 ohm	2J1-N	2J2-e
2S8	.2 ohm	2J2-Y	2J2-c
2S9	.2 ohm	2J1-d	2J2-c

3.5.6 Scale factor adjustments.- The scale factor adjustments of Drawing F10543700 sheet 4 shall be confirmed. 2R3 and 2R9 shall be within ± .001 of the ratio indicated. 2R10 shall be within ± .0005 of the ratio indicated. 2R4 shall be within ± .0015 of the ratio indicated.

3.5.7 Boresight function generator outputs.- The main gun and cupola gun elevation and deflection function generator output voltage ratios shall agree with the ratio settings listed in Table II for the knob positions given. The knob shall be set to the detent nearest the

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specified setting. The ratio tolerances listed in Table II include the equivalent to 0.05 mil maximum detent error.

TABLE II

KNOB SETTING	RATIO SETTING	RATIO TOLERANCE
Down or left 7 mils	.422222	± .001388
Down or left 3 mils	.466666	± .001388
Down or left 1 mil	.488888	± .001388
Zero	.500000	± .001388
Up or right 1 mil	.511111	± .001388
Up or right 3 mils	.533333	± .001388
Up or right 7 mils	.577777	± .001388

The main gun deflection and elevation function generator outputs at pins b and c respectively, of 2J1, shall be removed and the cupola gun functions generator outputs connected to these pins when pin F of 2J2 is connected to pin L of 2J1.

3.5.8 Range function generator output.- The output voltage ratios of the range function generator at pin d of 2J1 shall agree with the ratios listed in Table III.

TABLE III

RANGE KNOB SETTING	RATIO	RATION TOLERANCE
3975 meter	.420131	± .000393
3000 meter	.489657	± .000467
2500 meter	.535370	± .000550
2000 meter	.590302	± .000636
1500 meter	.657589	± .000754
1200 meter	.705825	± .000842
1000 meter	.742123	± .000912
500 meter	.851764	± .001137
325 meter	.898305	± .001241

3.5.9 Wind function generator output.- The output voltage ratio of the wind function generator at pin X of 2J1 shall agree with the ratios given in Table IV.

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TABLE IV

WIND DIAL SETTING	RATIO	RATIO TOLERANCE
Right 40 mph	.250000	± .002890
Right 20 mph	.375000	± .002265
Zero	.500000	± .000625
Left 20 mph	.625000	± .002265
Left 40 mph	.750000	± .002890

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract of purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) submission.- When required by the contract, an initial production sample shall be submitted to a Government approved facility for test and evaluation in accordance with the contract schedule. The sample shall consist of three units. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply. After inspection and provisional acceptance at source, the initial production sample shall be inspected at the Government approved facility for all requirements of the drawings and specifications. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

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4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents (IEL), when provided with the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection.- Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables V, VI, and VII herein. Examination and tests for packaging and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables V, VI, and VII shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejction by item or lot.

TABLE V - CLASSIFICATION OF DEFECTS

CLASS		REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE			
MAJOR: AQL 0.65% Defective			
101.	Vibration	3.3.3	4.6.1.1
102.	Continuity	3.4.1	Applicable Dwgs visual-ohmmeter
103.	Input power	3.5.1	4.6.2
104.	Power supply output	3.5.2	4.6.3
105.	Switch lens illumination	3.5.3	4.6.4
106.	Control indicator lamp	3.5.4	4.6.5
107.	Switch resistance	3.5.5	4.6.6
108.	Scale factor adjustments	3.5.6	4.6.7
109.	Boresight function generator output	3.5.7	4.6.8
110.	Range function generator output	3.5.8	4.6.9
111.	Wind function generator output	3.5.9	4.6.10
MINOR: NONE			

NOTE: Tests of Table V shall be conducted at a temperature between + 60°F and + 90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

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4.4 Special sampling.

4.4.1 General.- One control unit shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table VI.

TABLE VI

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301. Fabrication	3.1	Applicable Drawing - Visual
302. General specification	3.2	MIL-F-13926 - Visual

4.4.2 Environmental.- Three control units shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall have met the requirements and tests in Table VII.

TABLE VII

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303. Storage temperatures	3.3.1	4.6.1.2
304. Operating temperatures	3.3.2	4.6.1.3
305. Water resistance (sealing)	3.3.4	4.6.1.4

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with applicable technical data listed in the List of Inspection Documents, IEL 10543900.

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4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.3 Test equipment.- In conjunction with 4.5.2, the standard test equipment listed in Table VIII shall be utilized in performance of the applicable tests as specified in 4.6.

TABLE VIII

	NOMENCLATURE	DESCRIPTION
1.	A.C. Power supply	0-115 volts, adjustable, 400 cps ± 1%, 2% maximum harmonic distortion; variation in harmonic distortion shall not exceed 0.5%. Impedance less than 0.10 ohm from D.C. to 10 K.C.
2.	Oscilloscope	Sensitivity of 1 mv/cm.
3.	A.C. Ratio standard	6 Digit readout, to verify ratio settings as specified in 3.5.6, 3.5.7, 3.5.8, and 3.5.9 minimum input impedance, 500,000 ohms at 400 cps, output impedance less than 10 ohms.
4.	Resistance bridge	To test switch resistance as specified in 3.5.5.
5.	A.C. Voltmeter	True RMS reading accuracy of 1.0% of reading to measure output of 3.5.2.
6.	D.C. Voltmeter	± 0.10% accuracy.

4.6 Test methods and procedures.

4.6.1 Environmental.

4.6.1.1 Vibration.- To determine conformance to 3.3.3 the commander's control unit, shall be mounted on a suitable mounting apparatus and attached to a vibrating machine or fixture (see 4.5.2) and vibrated at the amplitude,

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frequency and time specified. After vibration, the unit shall be given a visual-tactile examination for damage and loose parts and scale movement. The units shall be subjected to a continuity test and the tests specified in 4.6.2 through 4.6.10 inclusive to determine conformance to the respective requirements referenced therein.

4.6.1.2 Storage temperatures.- To determine conformance to 3.3.1, the unit shall be tested in accordance with temperature cycling Procedure II of MIL-F-13926 for the specified temperatures. At the conclusion of these tests, the unit shall show no evidence of damage or deterioration when visually examined.

4.6.1.3 Operating temperatures.- To determine conformance to 3.3.2, the unit shall be placed in appropriate environmental chambers (see 4.5.2). While thermally stabilized at the specified temperatures, the unit shall be subjected to the tests specified in 4.6.2 through 4.6.10 inclusive to determine conformance with the requirements referenced therein.

4.6.1.4 Water resistance.- To determine conformance to 3.3.4, the unit shall be tested in accordance with Rain Procedure 1 of MIL-F-13926. The unit shall show no evidence of water damage as a result of this test when visually examined.

4.6.2 Input power.- To determine conformance to 3.5.1, when units are subjected to tests 4.6.3 through 4.6.10, the applicable test equipment items of 4.5.3 shall be checked to assure they provide proper power inputs to the terminals specified.

4.6.3 Power supply output.- To determine conformance to 3.4.2, with the a.c. power and load as specified, the d.c. voltage and ripple shall be checked with appropriate items from 4.5.2. The voltage, ripple, and ripple frequency shall be as specified.

4.6.4 Switch lens illumination.- To determine conformance to 3.5.3, with the a.c. supply as specified, the four lenses shall be observed for a change in intensity of illumination as the various pins are connected to the specified point. Rheostat 2R1 shall vary the illumination of the switch lenses.

4.6.5 Control indicator lamp.- To determine conformance to 3.5.4, pin Z of 2J1 shall be connected to pin A of 2J1 through 100 ohms and the override switch actuated. Each actuation of the switch shall cause the alternate control lamp to illuminate. Removal of the connection at pin A of 2J1 shall cause the red lamp only to illuminate. Rheostat 2R1 shall vary the illumination of the lamps.

4.6.6 Switch resistance.- To determine conformance to 3.5.5 switches listed on Table I shall be given a resistance check over the connections specified. A.C. power shall be removed during this test.

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4.6.7 Scale factor adjustments.- To determine conformance to 3.5.6, the adjustment of the specified potentiometers shall be checked. The test circuit of Drawing F10543700, sheet 4 shall be used. The ratios shall be within the tolerances specified.

4.6.8 Boresight function generator output.- To determine conformance to 3.5.7, the commander's control unit shall be inserted in the test circuit of Figure 1. The boresight knobs shall be set to the position specified in Table II and the output voltage ratios shall be within the tolerance specified. Connection of 2J2-F to 2J1-L shall remove the main gun deflection and elevation function generator output at pins b and c respectively, of 2J1 and connect the cupola gun function generator to these pins.

4.6.9 Range function generator output.- To determine conformance to 3.5.8, the unit shall be inserted in the test circuit of Figure 2. The range knob shall be set to ranges specified in Table III and the corresponding output voltage ratios shall be within the tolerances specified.

4.6.10 Wind function generator output.- To determine conformance to 3.5.9, after inserting the unit in the test circuit of Figure 3 the wind dial shall be set to the positions specified in Table IV and the corresponding output voltage ratio at the specified pin shall be within the tolerance specified.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with MIL-P-14232 and packaging data sheet MIL-P-14232/P10543700. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The commander's control unit is intended as a functional component of the electronic analog computer; Computer, Ballistics XM19.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of preservation, packaging and packing.
(c) Applicable packaging data sheet number (see 5.1).
(d) Applicable stock number.
(e) Order of document precedence.

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Preparing activity:

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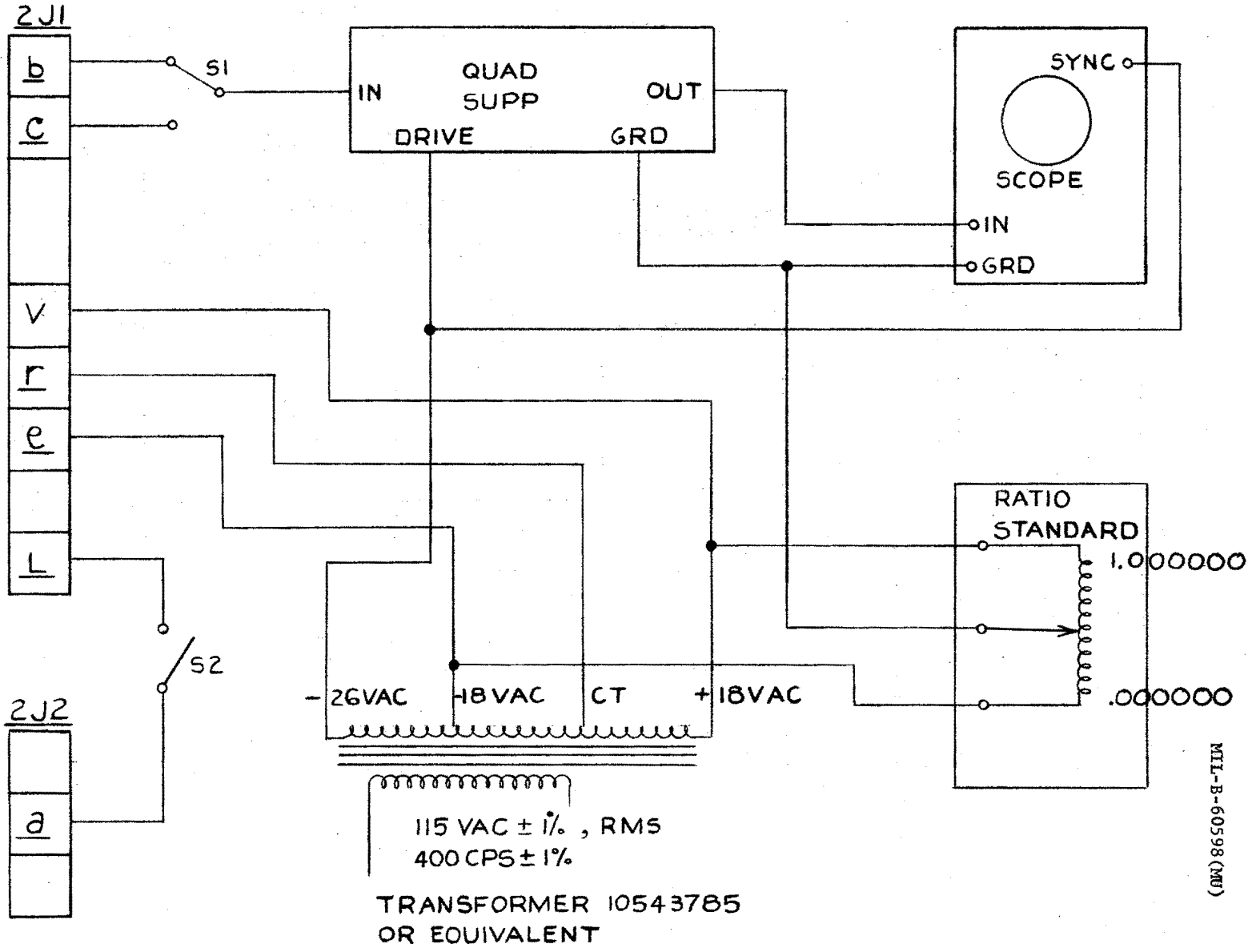


FIG.1 BORESIGHT FUNCTION GENERATORS

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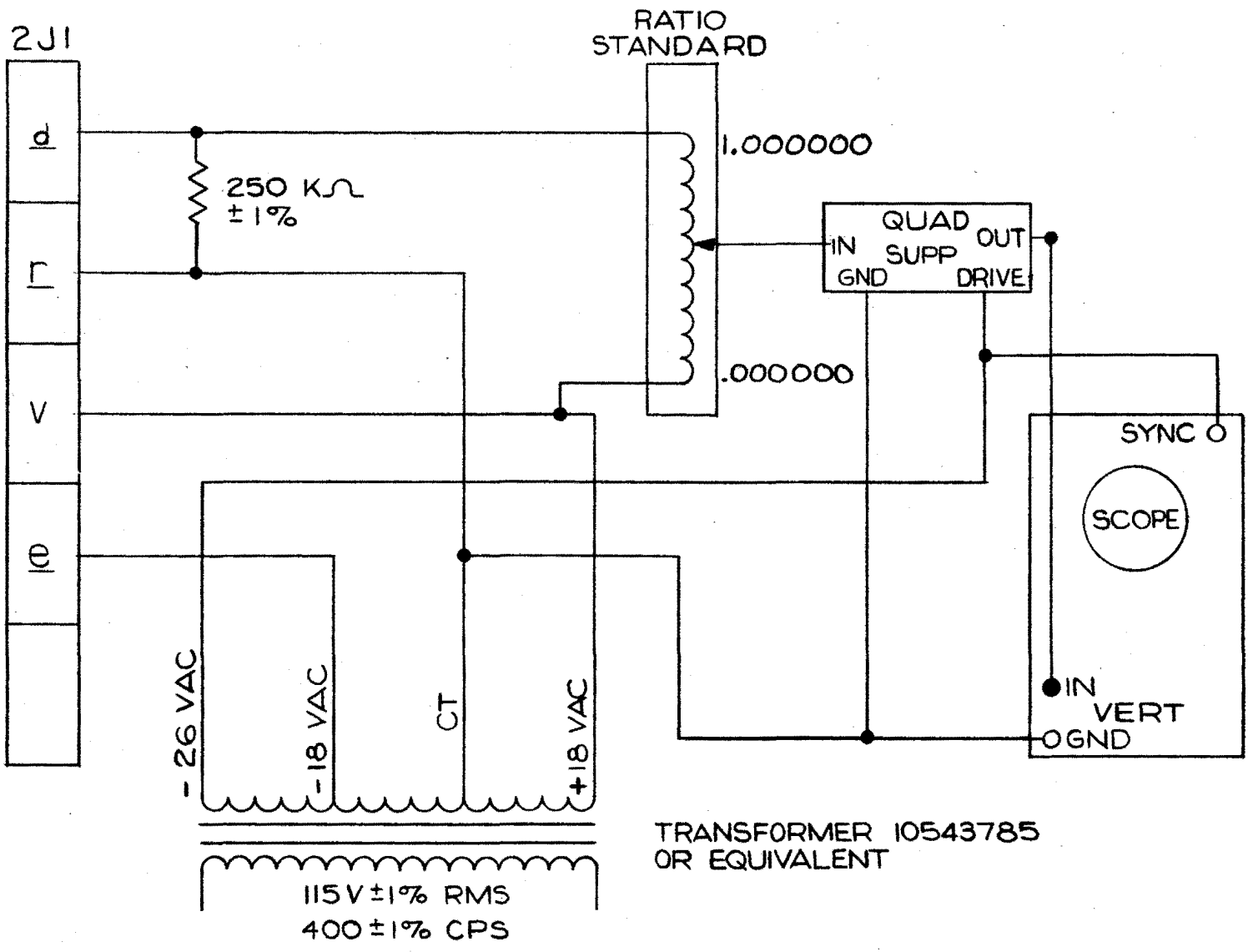


FIG. 2 RANGE OUTPUT

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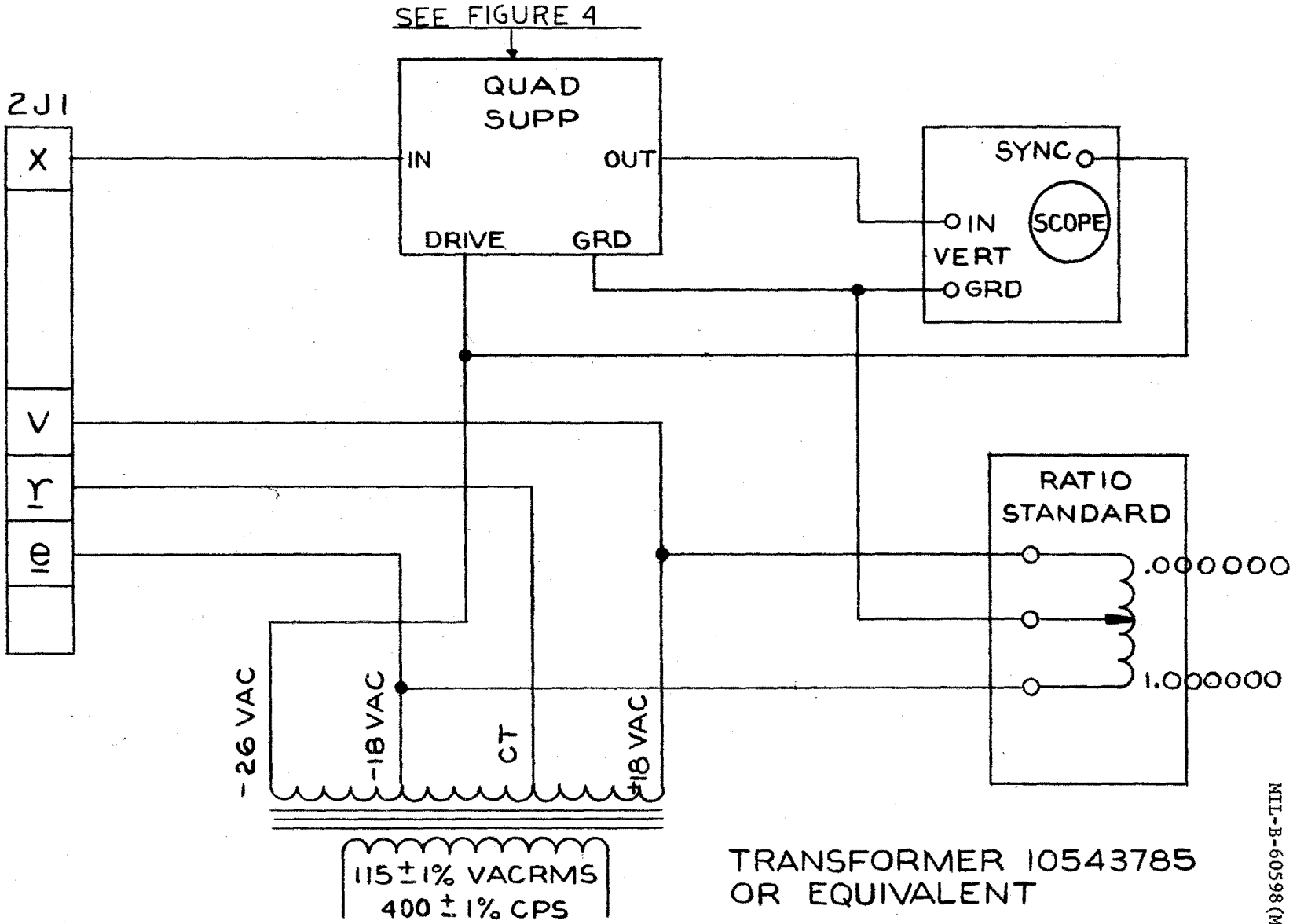


FIG. 3 WIND FUNCTION

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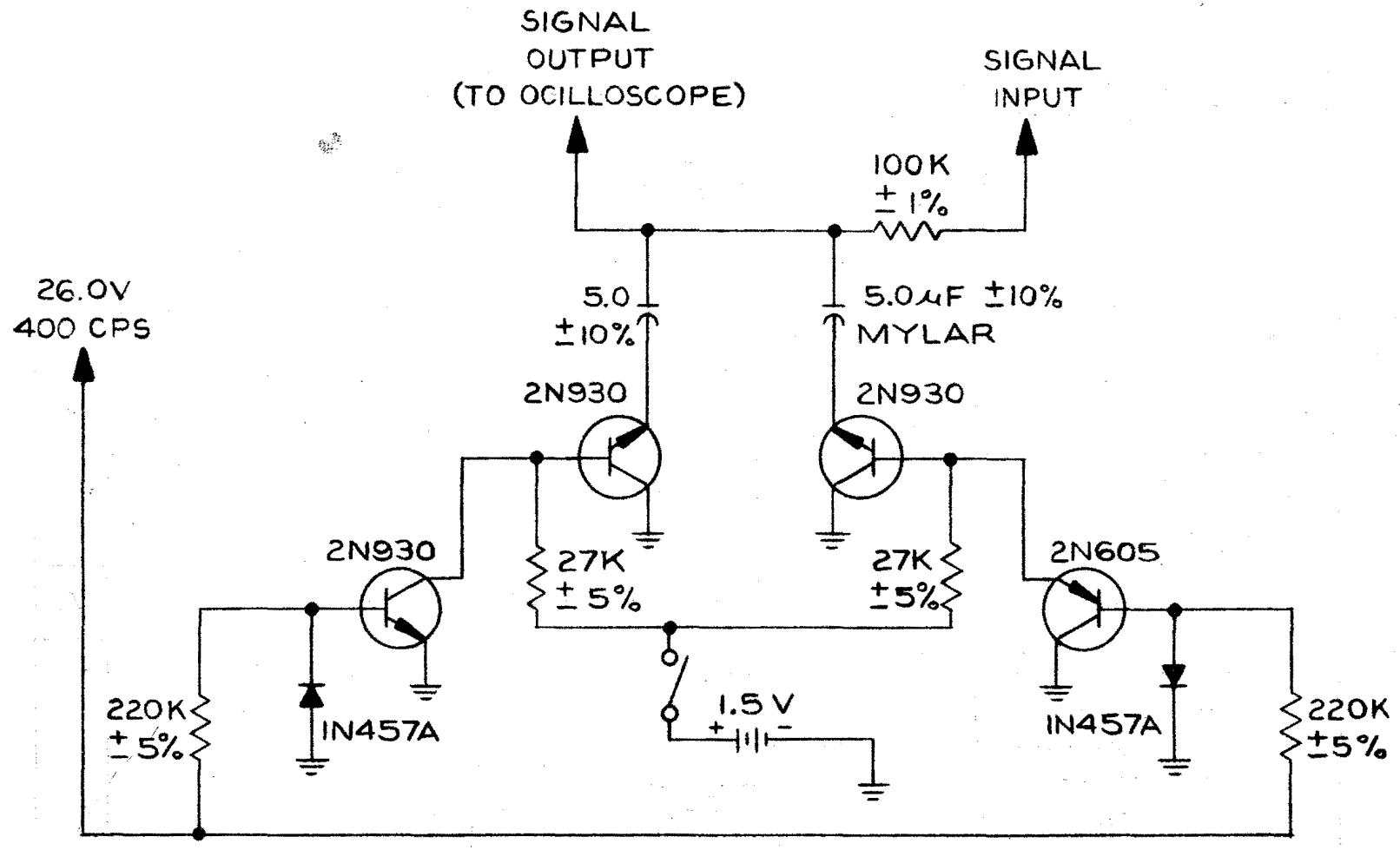


FIG. 4 QUADRATURE SUPPRESSOR

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